

Welcome!



**Have you ever felt
like your training
wasn't producing results?**



If yes! Meet Apex



Your AI-powered personal trainer



Apex, maximizing your training



- **Aaron Ricca**
 - Electrical and computer engineering
 - Ex professional skier
 - Climber
- **Derya Özsoy**
 - Informatik
 - Kickboxing



What's the problem ?

- training is far from optimized
- hard to find good exercises
- time is always too little



How does **Apex** work?

- the End User enters his **data**
- Apex learns with the collected **User data** and **data** from Internet and tries to find common similarities.
- At the end the User receives a personalized training plan



Data time !

Input attributes:

- Objective
- Available time
- Favorite exercises
- Biological attributes (Height, body fat%, medical condition, etc.)
- Alimentation (Cuisine, Diet type, enhancing substances, etc.)

Output attributes:

- Training periodization
- List of exercises
- Alimentation improvement tips
- Corresponding improvement % [Continuous]



Data Sources

Research & Literature

- Scientific papers
- Studies & publications
- Medical research

Real-World Data

- Gyms & fitness centers
- Professional coaches
- Testers & athletes

User-Generated

- Surveys & questionnaires
- Smartwatch data
- Fitness app tracking

Continuous Growth

- Data collected from users



Data Preprocessing

- Convert different units & scales to SI units
 - Height (feet -> cm), Weight (lb -> kg), etc.
- **Standardization** → for classification algorithms (KNN, Random Forest)
- **Normalization** → for regression algorithms



Few data?

-> More data?



Synthetic data!



SMOTE

Synthetic Minority Oversampling Technique

- Used for **imbalanced datasets**
→ e.g., 90% fit people and 10% overweight
- Problem: ML models **bias the majority class**
- SMOTE **creates new synthetic data** of the minority class



How SMOTE Works

Our scenario: 10% overweight, 90% fit

-**User A:** Age 35, Weight 95kg, Beginner level

-**User B:** Age 40, Weight 100kg, Beginner level

SMOTE finds two **similar overweight users** and creates a new realistic profile:

-**Synthetic user C:** Age 37, Weight 97kg, Beginner level ✨

✅ Makes the minority class **denser and more balanced** (27%)



What are we using for Apex

- **Supervised Learning**
 - trained using labeled training data
 - learns to recognize patterns in order to make predictions for new, unknown and similar data
 - All Datas are structured
 - Classification
 - Regression

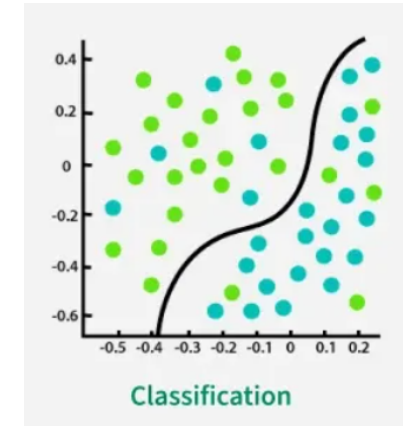


Tools

Classification

focuses on sorting observations into specific class

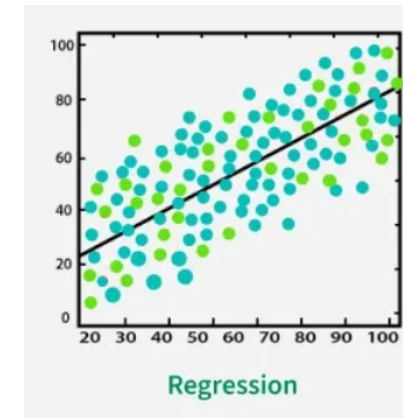
-> exp: We categorise some Data "sick" an "healthy" so that the user receives a training plan that does not harm his health



Regression 🎯

Based on making prediction or identification of trends

-> exp: predict a good Trainingsplan for the User



Algorithms

- **Linear Regression**
 - Predict improvement percentages
 - Estimate workout outcomes
- **K-Nearest Neighbors (KNN)**
 - Find similar user profiles
 - Recommend based on comparable cases
- **Random Forests**
 - Robust decision-making
 - Handles complex non-linear relationships
 - Gradient Boosting



Data Processing 🛠️

- **SMOTE** - Balance underrepresented groups
- **Cross-Validation** - Ensure model generalization
- **PCA (Principal component analysis)** - Reduce the data dimensionality



Why is **Apex** better?

- **Open source** - Transparent algorithms, community improvements
- **Data-driven optimization** - Personalized vs generic programs
- **Lower injury risk** - Biomechanical constraints and progressive overload
- **First of its kind** - Novel ML-based personal training approach
- **Continuous learning** - Improves with more user data



Possible risks, barriers and obstacles

- **Incomplete data can lead to incorrect results.**
 - can lead to injuries due to incorrect training suggestions
- **Time and resources.**
 - Data must always be updated and maintained.
 - Possible data loss.
 - High costs und delays.
- **Data breach.**



Ethical Considerations

- **Privacy & GDPR compliance** - Sensitive biometric and health data
- **Labor market impact** - Potential displacement of personal trainers
- **Economic effects** - Reduced gym memberships
- **Algorithmic bias** - Training data skewed toward young, fit males
 - Risk: Inappropriate recommendations for underrepresented groups
 - Can lead to injuries in elderly or different body types
- **Dual-use concerns** - Technology repurposed beyond original intent
 - Example: Military training



Sources



